



the AI antenna engineer

EVERY WIRELESS THING

If it connects, it has an antenna.

5.4B

Bluetooth devices a year

3.8B

Wi-Fi devices a year

6–12

antennas in one phone

A FULL-TIME SPECIALISM

An RF engineer places it by hand, one guess at a time.

01 GUESS Pick a corner JUDGMENT & EXPERIENCE	02 MODEL Mesh it in HFSS HOURS OF SETUP	03 WAIT Full-wave solve ≈ 1 HOUR – ONE CANDIDATE	04 TUNE Build & measure DAYS TO WEEKS
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DOZENS OF TIMES, BEFORE SIGN-OFF

≈ 1 h

ONE FULL-WAVE RUN OF A WHOLE DEVICE

≈ 41 h

SOLVER TIME TO TRY A REAL SET OF POSITIONS

1 wk

ONE RF ENGINEER ON THE CALENDAR

BIG, AND COMPOUNDING

\$28B of antennas ship a year. It doubles by 2035.

SEGMENT	NOW	FORECAST
Antenna hardware	\$28B	\$54B '35
5G antennas	\$23B	\$50B '30
Design & simulation software	\$1.5B	\$2.9B '34

WHAT IS PUSHING IT	
5G, and then 6G	+16.9% / yr
IoT endpoints by 2030	30B+
Antennas per car by 2030	4 → 8

YOU HAVE SEEN THIS BEFORE

Software had this exact loop. Then AI ate it.

SOFTWARE, 2021

- / 47M developers
- / \$7.6B of tools, none of them with judgment

SOFTWARE, NOW

- / \$3.5B of AI coding tools, from zero in three years
- / 84% of developers use them

Antenna design is the same loop. Nobody has built the agent for it.

TWO INPUTS, ONE LOOP

Drop in the build file and the spec. Kevin does the rest.

01 UPLOAD .blend + spec SECONDS	02 READ Geometry & keep-outs 191 PARTS, NO SOLVER	03 PROPOSE Design & placement WITH ITS REASONING	04 SIMULATE Score each candidate 83 MS – ONE CANDIDATE	05 ITERATE Until it converges IT DECIDES WHEN TO STOP
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FULLY AUTONOMOUS – NO HUMAN IN THE LOOP

38

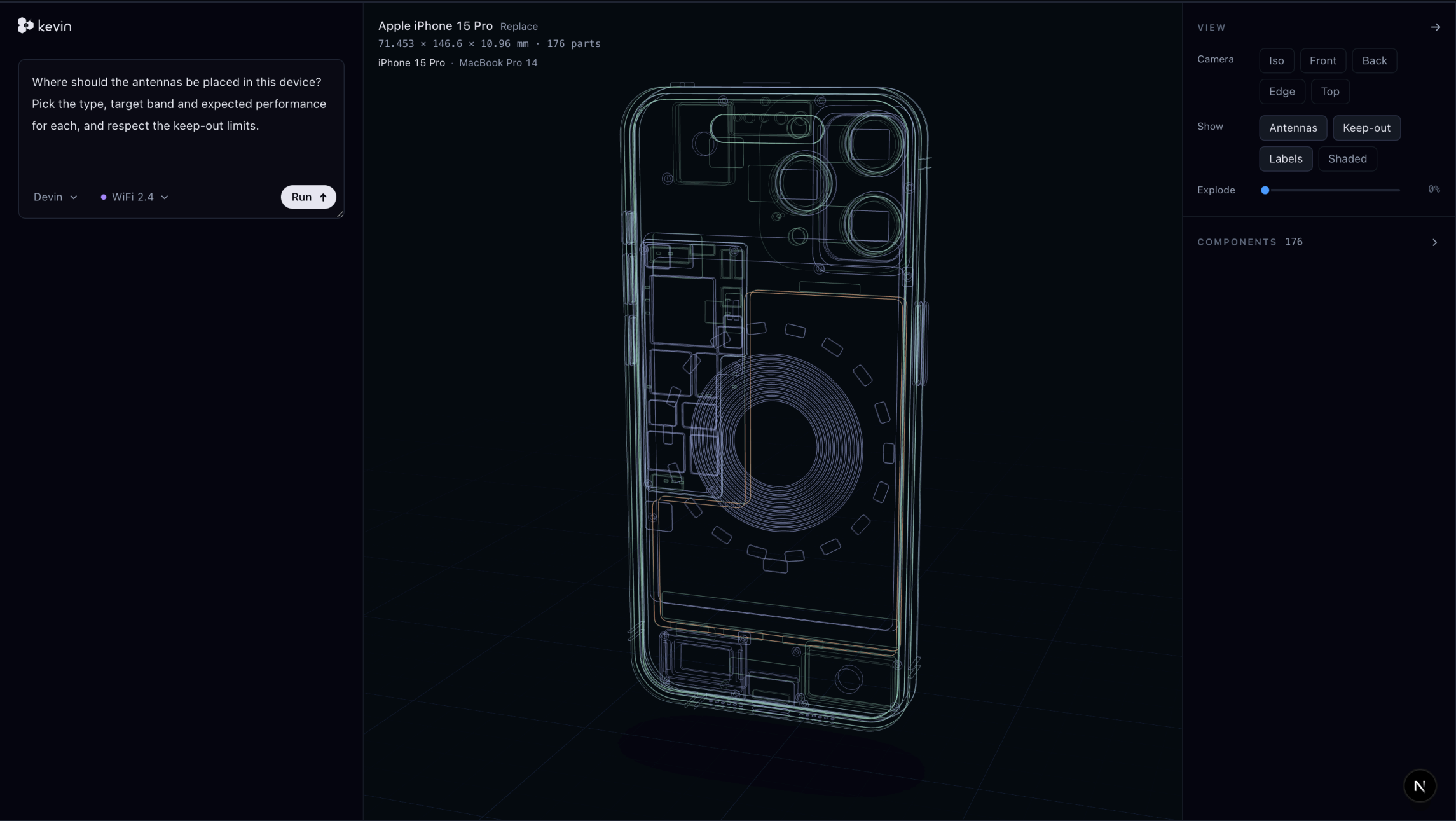
CANDIDATES SIMULATED

1 min 29 s

UPLOAD TO FINISHED DESIGN

0

ENGINEER HOURS WAITING ON A SOLVER



LIVE

Let's run one.

- / It reads the real device
- / It argues out loud
- / It stops on its own



**Aleksandr
Gorbunov**

Agent & systems integration



**Damià
Vicens Ramis**

3D modelling & simulation



**Amr
Moussa**

Visualisation & UI



**Rauf
Boudenne**

RF research & simulation

questions.

Build file in. Defensible placement out.